MTH129 Introduction to AI for Mathematical Sciences (MS)

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SMP AI Team

Linear Algebra

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Introduction of Python

- Python is created by Guido van Rossum in 1980-1990
- A program
  - Compiled Languages: C/C++, Fortran, Java, ...
  - Interpreted Languages: Python, R, MATLAB, ...
  - Python interpreters: python, ipython, jupyter
- A programming language
  - General-purpose
  - Object-oriented (everything in Python is an object)
  - Dynamically typed
  - Interpreted (each statement executed sequentially, and hence are typically slower than for compiled languages)
  - Extensible
- A software ecosystem
  - Python Standard Library
  - Third-party packages
  - NumPy: Numpy is the fundamental package for scientific computing in Python. We can “import numpy as np” to import the package.

Fundamental Operations

Operations

- addition: +
- substraction: -
- multiplication: *
- division: /
- power: **
- modulo: %
- floor division: //

Example

```
(((6+5)-4)*3/2)**2%4
```

2.25

1.1 Review

- A matrix $\mathbf{A}$ is a system of numbers with n rows and p columns:

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & \cdots & \cdots & a_{1p} \\ \vdots & a_{22} & & & & \vdots \\ \vdots & \vdots & \ddots & & & \vdots \\ \vdots & \vdots & & \ddots & & \vdots \\ \vdots & \vdots & & & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & \cdots & \cdots & a_{np} \end{pmatrix}$$

- We also write (a_{ij}) for $\mathbf{A}$ and $\mathbf{A}_{(n\times p)}$ to indicate the numbers of rows and columns.
- The size of a matrix is the number of rows by the number of columns. For example, we say the size of $\mathbf{A}_{(n\times p)}$ is n by (times) p.
- Vectors are matrices with one column and are denoted as $\mathbf{x}$ or $\mathbf{x}_{(p\times 1)}$.

Elementary Operations

- $\mathbf{A}^T = (a_{ji})$, transpose of $\mathbf{A}$.
  Python code: np.transpose(A) or A.T (Here, np is the abbreviation of numpy.)
- Let $\mathbf{B}$ has the same size as $\mathbf{A}$
  - $\mathbf{A} + \mathbf{B} = (a_{ij} + b_{ij})$.
    Python code: $A + B$ or np.add(A, B).
  - $\mathbf{A} - \mathbf{B} = (a_{ij} - b_{ij})$.
    Python code: $A - B$ or np.subtract(A, B).
- For a scalar c, $c \cdot \mathbf{A} = (c \cdot a_{ij})$.
  Python code: c * A or np.multiply(c, A).
- Matrix production:
  $\mathbf{A} \cdot \mathbf{B} = \mathbf{A}_{(n\times p)}\mathbf{B}_{(p\times m)} = \mathbf{C}_{(n\times m)} = (c_{ik}) = \left(\sum_{j=1}^{p} a_{ij}b_{jk}\right)$.
  Python code: A@B or np.dot(A, B).
  Remark. For two matrices A and B, np.multiply(A, B) is also well defined. But the result is not the product of matrix multiplication, but that of the element-wise multiplication.

For conformable matrixes (a matrix is conformable if its dimensions are suitable for defining some operations), we have

- $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$
- $\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}$
- $\mathbf{A}(\mathbf{BC}) = (\mathbf{AB})\mathbf{C}$
- $\left(\mathbf{A}^T\right)^T = \mathbf{A}$
- $(\mathbf{AB})^T = \mathbf{B}^T\mathbf{A}^T$

Properties and Definitions

- Linearly Independent: A set of p vectors $\mathbf{x}_1, \cdots, \mathbf{x}_p$ are said to be linearly independent if $\sum_{j=1}^{p} c_j \mathbf{x}_j = \mathbf{0}$ implies $c_j = 0, \forall j$, where $c_1, \ldots, c_p$ are scalars.
- Rank: The rank of a matrix $\mathbf{A}_{(n \times p)}$ is defined as the maximum number of linearly independent rows (columns).
  Python code: np.linalg.matrix_rank(A)
- Trace: The trace of a square matrix $\mathbf{A}_{(p \times p)}$ is the sum of its diagonal elements $\sum_{i=1}^{p} a_{ii}$.
  Python code: np.trace(A)

Determinant

Determinant:
For a square matrix $\mathbf{A}$, it is defined as $|\mathbf{A}| = \sum(-1)^{|\tau|} a_{1\tau(1)} \dots a_{p\tau(p)}$, the summation is over all permutations τ of $\{1, 2, \dots, p\}$, and $|\tau| = 0$ if the permutation can be written as a product of an even number of transpositions and $|\tau| = 1$ otherwise.

- For a 2×2 matrix $\mathbf{A}$, $|\mathbf{A}| = a_{11}a_{22} - a_{12}a_{21}$.
- Python code: np.linalg.det(A)
- Properties

$$\left|\mathbf{A}^T\right| = |\mathbf{A}|$$
$$|\mathbf{AB}| = |\mathbf{A}| \cdot |\mathbf{B}|$$
$$|c\mathbf{A}| = c^n|\mathbf{A}|.$$

Inverse Matrix

- Inverse: If $|\mathbf{A}_{(p\times p)}| \neq 0$, then the inverse $\mathbf{A}^{-1}$ exists and satisfies

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}_p,$$

  where $\mathbf{I}_p$ is the identity matrix with ones on the main diagonal and zeros elsewhere.
  - For a 2×2 matrix $\mathbf{A}$, $\mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|}\begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}$.
  - Python code: np.linalg.inv(A).
- Generalized Inverse (Reading Material): A more general concept is the G-inverse (Generalized Inverse) $\mathbf{A}^{-1}$ which satisfies the following

$$\mathbf{A}\mathbf{A}^{-1}\mathbf{A} = \mathbf{A}.$$

Eigenvalues and Eigenvectors

Let $\mathbf{A}_{(p\times p)}$

- If there exists a scalar λ and a vector $\mathbf{x}$ satisfying

$$\mathbf{A}\mathbf{x} = \lambda\mathbf{x},$$

  then we call $(\lambda, \mathbf{x})$ is a pair of eigenvalue and eigenvector.
- An eigenvalue λ is a root of $|\mathbf{A} - \lambda\mathbf{I}_p| = 0$.
- Python code: np.linalg.eigvals(A) or np.linalg.eig(A).
- There are up to p eigenvalues $\lambda_1, \ldots, \lambda_p$ (including multiplicity). For each eigenvalue λ_j, a corresponding eigenvector $\mathbf{x}_j$ exists.
- The determinant and trace of $\mathbf{A}$ can be rewritten as

$$|\mathbf{A}| = |\mathbf{D}| = \prod_{j=1}^{p} \lambda_j \text{ and } \operatorname{tr}(\mathbf{A}) = \operatorname{tr}(\mathbf{D}) = \sum_{j=1}^{p} \lambda_j.$$

Diagonalization

If $\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_p$ are linearly independent, we then say $\mathbf{A}$ is diagonalizable and have

$$\mathbf{A} = \mathbf{PDP}^{-1},$$

where $\mathbf{P} = [\mathbf{x}_1 \ \mathbf{x}_2 \ \cdots \ \mathbf{x}_p]$ and $\mathbf{D} = \mathrm{diag}\,(\lambda_1, \ldots, \lambda_p)$ is a diagonal matrix with $d_{jj} = \lambda_j$ and 0 otherwise.

Example 1

Let us consider the matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix}.$$

Find **AA**, the rank, trace, transpose, determinant, inverse, eigenvalues and the associated eigenvectors of **A**.

Solution:

$\mathbf{AA} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix} = \mathbf{A}$. If $\mathbf{AA} = \mathbf{A}$, we then say **A** is **idempotent**.

Rank is 2.

Trace is $1 + (1/2) + (1/2) = 2$.

$\mathbf{A}' = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix} = \mathbf{A}$. If $\mathbf{A}' = \mathbf{A}$, then we say **A** is **symmetric**.

$|\mathbf{A}| = 0$ as two rows are identical. Actually $|\mathbf{A}_{(p\times p)}| \neq 0 \Leftrightarrow$ rank $= p$, and $|\mathbf{A}_{(p\times p)}| = 0 \Leftrightarrow$ rank $< p$.

Since $|\mathbf{A}| = 0$, $\mathbf{A}$ is not invertible.

$$|\mathbf{A} - \lambda I_p| = \left| \begin{bmatrix} 1-\lambda & 0 & 0 \\ 0 & \frac{1}{2}-\lambda & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2}-\lambda \end{bmatrix} \right| = (1-\lambda)[(1/4 - \lambda + \lambda^2) - 1/4]$$
$$= -\lambda(\lambda - 1)^2,$$

suggests that $\lambda_1 = 0$, $\lambda_2 = 1$ and $\lambda_3 = 1$.

As expected, we have $\sum_{i=1}^{3} \lambda_i = 0 + 1 + 1 = 2$ as the trace of $\mathbf{A}$ and $\prod_{i=1}^{3} \lambda_i = (0)(1)(1) = 0$ as the determinant of $\mathbf{A}$.

For $\lambda_1 = 0$, $\mathbf{Ax} = \lambda_1 \mathbf{x} = \mathbf{0}$ suggests

$$\begin{bmatrix} x_1 \\ (1/2)x_2 + (1/2)x_3 \\ (1/2)x_2 + (1/2)x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Hence $\mathbf{x}_1' = [0 \quad 1/\sqrt{2} \quad -1/\sqrt{2}]$ (we normalize the magnitude (length) of $\mathbf{x}_1$ to 1).
For $\lambda_2 = \lambda_3 = 1$, $\mathbf{Ax} = \mathbf{x}$ suggests

$$\begin{bmatrix} x_1 \\ (1/2)x_2 + (1/2)x_3 \\ (1/2)x_2 + (1/2)x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}.$$

Hence $\mathbf{x}_2' = [1 \quad 0 \quad 0]$ and $\mathbf{x}_3' = [0 \quad 1/\sqrt{2} \quad 1/\sqrt{2}]$ (solutions are not unique).

Let

$$\mathbf{P} = \begin{bmatrix} 0 & 0 & 1 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ -1/\sqrt{2} & 1/\sqrt{2} & 0 \end{bmatrix} \text{ and } \mathbf{D} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

we have

$$\mathbf{P}^{-1} = \begin{bmatrix} 0 & 1/\sqrt{2} & -1/\sqrt{2} \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ 1 & 0 & 0 \end{bmatrix} \text{ and } \mathbf{PDP}^{-1} = \mathbf{A}.$$

1.2 The Gram-Schmidt Process

Introduction

- Two vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}$ are **orthogonal** if $\mathbf{u}^T\mathbf{v} = 0$ (or $\mathbf{u} \cdot \mathbf{v} = 0$).
  - $\mathbf{u} = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$, and $\mathbf{v} = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$ are orthogonal.
- An orthogonal projection: Given a nonzero vector $\mathbf{u}$ in $\mathbb{R}^n$, a vector $\mathbf{y}$ in $\mathbb{R}^n$ could be decomposed into the sum of two vectors, one a multiple of $\mathbf{u}$ and the other orthogonal to $\mathbf{u}$, i.e.,

$$\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z},$$

where $\hat{\mathbf{y}}$ is called the orthogonal projection of $\mathbf{y}$ onto $\mathbf{u}$, and

$$\hat{\mathbf{y}} = \frac{\mathbf{y}^T\mathbf{u}}{\mathbf{u}^T\mathbf{u}}\mathbf{u}$$

  - Let $\mathbf{y} = \begin{bmatrix} 7 \\ 6 \end{bmatrix}$ and $\mathbf{u} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$, then $\hat{\mathbf{y}} = \frac{\mathbf{y}^T\mathbf{u}}{\mathbf{u}^T\mathbf{u}}\mathbf{u} = \frac{40}{20}\mathbf{u} = 2\mathbf{u}$.

The Orthogonal Decomposition Theorem

Let $\mathbf{W}$ be a subspace of $\mathbb{R}^n$. Then each $\mathbf{y}$ in $\mathbb{R}^n$ can be written uniquely in the form

$$\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z},$$

where $\hat{\mathbf{y}}$ is in $\mathbf{W}$ and $\mathbf{z}$ is in $\mathbf{W}^{\perp}$. In fact, if $\{\mathbf{u}_1, \cdots, \mathbf{u}_p\}$ is any orthogonal basis of $\mathbf{W}$, then

$$\hat{\mathbf{y}} = \frac{\mathbf{y}^T\mathbf{u}_1}{\mathbf{u}_1^T\mathbf{u}_1}\mathbf{u}_1 + \cdots + \frac{\mathbf{y}^T\mathbf{u}_p}{\mathbf{u}_p^T\mathbf{u}_p}\mathbf{u}_p$$

and

$$\mathbf{z} = \mathbf{y} - \hat{\mathbf{y}}.$$

Note that $\mathbf{z} \perp \mathbf{u}_i$ for all $i = 1, \cdots, p$, i.e., $\mathbf{z}^T\mathbf{u}_i = 0$

The Gram-Schmidt Process

Given a basis $\{\mathbf{x}_1, \cdots, \mathbf{x}_p\}$ for a nonzero subspace $\mathbf{W}$ of $\mathbb{R}^n$, define

$$\begin{aligned}
\mathbf{v}_1 &= \mathbf{x}_1 \\
\mathbf{v}_2 &= \mathbf{x}_2 - \frac{\mathbf{x}_2^T\mathbf{v}_1}{\mathbf{v}_1^T\mathbf{v}_1}\mathbf{v}_1 \\
\mathbf{v}_3 &= \mathbf{x}_3 - \frac{\mathbf{x}_3^T\mathbf{v}_1}{\mathbf{v}_1^T\mathbf{v}_1}\mathbf{v}_1 - \frac{\mathbf{x}_3^T\mathbf{v}_2}{\mathbf{v}_2^T\mathbf{v}_2}\mathbf{v}_2 \\
&\vdots \\
\mathbf{v}_p &= \mathbf{x}_p - \frac{\mathbf{x}_p^T\mathbf{v}_1}{\mathbf{v}_1^T\mathbf{v}_1}\mathbf{v}_1 - \frac{\mathbf{x}_p^T\mathbf{v}_2}{\mathbf{v}_2^T\mathbf{v}_2}\mathbf{v}_2 - \cdots - \frac{\mathbf{x}_p^T\mathbf{v}_{p-1}}{\mathbf{v}_{p-1}^T\mathbf{v}_{p-1}}\mathbf{v}_{p-1}
\end{aligned}$$

Then $\{\mathbf{v}_1, \cdots, \mathbf{v}_p\}$ is an orthogonal basis for $\mathbf{W}$.

Remark: Orthonormal Bases (Orthogonal basis & unit magnitude)
Define $\mathbf{u}_i = \frac{1}{||\mathbf{v}_i||}\mathbf{v}_i$, where $||\mathbf{v}_i|| = \mathbf{v}_i^T\mathbf{v}_i$ is the length of the vector $\mathbf{v}_i$.
Then $\{\mathbf{u}_1, \cdots, \mathbf{u}_p\}$ is an Orthonormal basis for $\mathbf{W}$.

Example 2

Let $\mathbf{W} = \text{Span}\{\mathbf{x}_1, \mathbf{x}_2\}$, where $\mathbf{x}_1 = \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}$ and $\mathbf{x}_2 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$. Construct an orthonormal basis $\{\mathbf{u}_1, \mathbf{u}_2\}$.

Solution: $\mathbf{v}_1 = \mathbf{x}_1$

$$\mathbf{v}_2 = \mathbf{x}_2 - \frac{\mathbf{x}_2^T \mathbf{v}_1}{\mathbf{v}_1^T \mathbf{v}_1} \mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} - \frac{15}{45} \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

Then $\mathbf{u}_1 = \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \\ 0 \end{bmatrix}$ and $\mathbf{u}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

1.3 Diagonaliztion of Symmetric Matrics

Introduction

- In Example 1, we have $\mathbf{P}^{-1} = \mathbf{P}^T$. That is to say, $\mathbf{A} = \mathbf{PDP}^{-1} = \mathbf{PDP}^T$. What kind of matrix has this property?
- If $\mathbf{A}$ is symmetric, then any two eigenvectors that correspond to distinct eigenvalues are orthogonal.
  - **Proof:** Let $\mathbf{v}_1$ and $\mathbf{v}_2$ be the eigenvectors that correspond to distinct eigenvalues, say, λ_1 and λ_2.

$$\begin{aligned}\lambda_1 \mathbf{v}_1^T \mathbf{v}_2 =& (\lambda_1 \mathbf{v}_1)^T \mathbf{v}_2 = (\mathbf{A}\mathbf{v}_1)^T \mathbf{v}_2 \\ =& \mathbf{v}_1^T \mathbf{A}^T \mathbf{v}_2 = \mathbf{v}_1^T (\mathbf{A}^T \mathbf{v}_2) = \mathbf{v}_1^T (\lambda_2 \mathbf{v}_2) = \lambda_2 \mathbf{v}_1^T \mathbf{v}_2\end{aligned}$$

    Hence $(\lambda_1 - \lambda_2)\mathbf{v}_1^T \mathbf{v}_2 = 0$. $(\lambda_1 - \lambda_2) \neq 0$ suggests that $\mathbf{v}_1^T \mathbf{v}_2 = 0$, i.e., orthogonal.
- An $n \times n$ matrix $\mathbf{A}$ is said to be **orthogonally diagonalizable** if there are an orthogonal matrix $\mathbf{P}$ (with $\mathbf{P}^{-1} = \mathbf{P}^T$) and a diagonal matrix $\mathbf{D}$ such that

$$\mathbf{A} = \mathbf{PDP}^{-1} = \mathbf{PDP}^T$$

- An $n \times n$ matrix $\mathbf{A}$ is orthogonally diagonalizable if and only if $\mathbf{A}$ is a symmetric matrix.
  - $\mathbf{A}^T = (\mathbf{PDP}^T)^T = \mathbf{PD}^T\mathbf{P}^T = \mathbf{PDP}^T = \mathbf{A}$.
- **The Spectral Theorem for Symmetric Matrices**: An $n \times n$ matrix $\mathbf{A}$ has the following properties:
  1. $\mathbf{A}$ has n real eigenvalues, counting multiplicities.
  2. The dimension of the eigenspace for each eigenvalue λ equals the multiplicity of λ as a root of the characteristic equation.
  3. The eigenspaces are mutually orthogonal, in the sense that eigenvectors corresponding to different eigenvalues are orthogonal.
  4. $\mathbf{A}$ is orthogonally diagonalizable.
- **Spectral Decomposition**:

$$\mathbf{A} = \lambda_1 \mathbf{u}_1 \mathbf{u}_1^T + \cdots + \lambda_n \mathbf{u}_n \mathbf{u}_n^T$$

$$\begin{aligned}\mathbf{A} =& \mathbf{PDP}^T \\ =& [\mathbf{u}_1 \ \cdots \ \mathbf{u}_n] \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} \begin{bmatrix} \mathbf{u}_1^T \\ \vdots \\ \mathbf{u}_n^T \end{bmatrix} = [\lambda_1 \mathbf{u}_1 \ \cdots \ \lambda_n \mathbf{u}_n] \begin{bmatrix} \mathbf{u}_1^T \\ \vdots \\ \mathbf{u}_n^T \end{bmatrix}\end{aligned}$$

Example 3

Orthogonally diagonalize the matrix $\mathbf{A} = \begin{bmatrix} 7 & 2 \\ 2 & 4 \end{bmatrix}$, whose eigenvectors are $\mathbf{u}_1 = \begin{bmatrix} 2/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix}$ and $\mathbf{u}_2 = \begin{bmatrix} -1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix}$ that correspond to the eigenvalues $\lambda_1 = 8$ and $\lambda_2 = 3$.

1. Construct $\mathbf{P}$ and $\mathbf{D}$ such that $\mathbf{A} = \mathbf{PDP}^T$ with $\mathbf{P}^T = \mathbf{P}^{-1}$.
2. Construct a spectral decomposition.

Solution:

1. $$\mathbf{P} = \begin{bmatrix} 2/\sqrt{5} & -1/\sqrt{5} \\ 1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix}, \text{ and } \mathbf{D} = \begin{bmatrix} 8 & 0 \\ 0 & 2 \end{bmatrix}$$

2. $$8\mathbf{u}_1\mathbf{u}_1^T + 3\mathbf{u}_2\mathbf{u}_2^T = 8\begin{bmatrix} 4/5 & 2/5 \\ 2/5 & 1/5 \end{bmatrix} + 3\begin{bmatrix} 1/5 & -2/5 \\ -2/5 & 4/5 \end{bmatrix} = \mathbf{A}$$

1.4 The Singular Value Decomposition

Introduction

- Not all matrices can be factored as $\mathbf{A} = \mathbf{PDP}^{-1}$ with $\mathbf{D}$ diagonal.
- A factorization $\mathbf{A} = \mathbf{QDP}^{-1}$ is possible for any $m \times n$ matrix $\mathbf{A}$.
- The singular value decomposition (SVD), a special factorization of the above type, is one of the most useful matrix factorizations.
- The absolute values of the eigenvalues of a symmetric matrix $\mathbf{S}$ measure the amounts that $\mathbf{S}$ stretches or shrinks the eigenvectors. If $\mathbf{Sx} = \lambda\mathbf{x}$ and $||x|| = 1$, then

$$||\mathbf{Sx}|| = ||\lambda\mathbf{x}|| = |\lambda|||x|| = |\lambda|.$$

If λ_1 is the eigenvalue with the greatest magnitude, then a corresponding unit eigenvector $\mathbf{v}_1$ identifies a direction in which the stretching effect of $\mathbf{S}$ is greatest. That is, the length of $\mathbf{Sx}$ is maximized when $\mathbf{x} = \mathbf{v}_1$, and $||\mathbf{Sv}_1|| = |\lambda_1|$.

Example 4

If $\mathbf{A} = \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix}$, then the linear transformation $\mathbf{x} \mapsto \mathbf{Ax}$ maps the unit sphere $\{\mathbf{x} : ||\mathbf{x}|| = 1\}$ in $\mathbb{R}^3$ onto an ellipse in $\mathbb{R}^2$. Find a unit vector $\mathbf{x}$ at which the length $||\mathbf{Ax}||$ is maximized.

Solution: It is equivalent to maximize $||\mathbf{Ax}||^2$. Note that

$$||\mathbf{Ax}||^2 = (\mathbf{Ax})^T(\mathbf{Ax}) = \mathbf{x}^T\mathbf{A}^T\mathbf{Ax} = \mathbf{x}^T(\mathbf{A}^T\mathbf{A})\mathbf{x}.$$

Clearly, $\mathbf{A}^T\mathbf{A}$ is a square matrix, and actually

$$\mathbf{A}^T\mathbf{A} = \begin{bmatrix} 80 & 100 & 40 \\ 100 & 170 & 140 \\ 40 & 140 & 200 \end{bmatrix}.$$

Its eigenvalues are $\lambda_1 = 360$, $\lambda_2 = 90$, and $\lambda_3 = 0$. The unit eigenvectors are

$$\mathbf{v}_1 = \begin{bmatrix} 1/3 \\ 2/3 \\ 2/3 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} -2/3 \\ -1/3 \\ 2/3 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 2/3 \\ -2/3 \\ 1/3 \end{bmatrix},$$

For $||\mathbf{x}|| = 1$, the maximum value of $||\mathbf{Ax}||$ is $||\mathbf{Av}_1|| = \sqrt{\lambda_1} = 6\sqrt{10}$.

Singular Value

- Let $\mathbf{A}$ be an $m \times n$ matrix. Then $\mathbf{A}^T\mathbf{A}$ is symmetric and can be orthogonally diagonalized.
- Let $\{\mathbf{v}_1, \cdots, \mathbf{v}_n\}$ be an orthonormal (orthogonal and has unit length) basis for $\mathbb{R}^n$ consisting of eigenvectors of $\mathbf{A}^T\mathbf{A}$, and let $\lambda_1, \cdots, \lambda_n$ be the associated eigenvalues of $\mathbf{A}^T\mathbf{A}$.
- Since

$$||\mathbf{A}\mathbf{v}_i||^2 = (\mathbf{A}\mathbf{v}_i)^T(\mathbf{A}\mathbf{v}_i) = \mathbf{v}_i^T(\mathbf{A}^T\mathbf{A})\mathbf{v}_i = \mathbf{v}_i^T\lambda_i\mathbf{v}_i = \lambda_i,$$

  suggesting that the eigenvalues of $\mathbf{A}^T\mathbf{A}$ is nonnegative.
- Without losing of any generality, we may assume that the eigenvalues are arranged so that

$$\lambda_1 \geq \lambda_2 \cdots \geq \lambda_n \geq 0$$

- The **singular values** of $\mathbf{A}$ are the square roots of the eigenvalues of $\mathbf{A}^T\mathbf{A}$, denoted by $\sigma_1, \cdots, \sigma_n$, where $\sigma_i = \sqrt{\lambda_i}$ for $1 \leq i \leq n$. The singular values of $\mathbf{A}$ are the lengths of the vectors $\mathbf{A}\mathbf{v}_1, \cdots, \mathbf{A}\mathbf{v}_n$.

The Singular Value Decomposition

Let $\mathbf{A}$ be an $m \times n$ matrix with rank r. Then there exists an $m \times n$ matrix $\mathbf{\Sigma}$ as in the form

$$\mathbf{\Sigma} = \begin{bmatrix} \mathbf{D} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix},$$

for which the diagonal entries in $\mathbf{D}$ are the first r singular values of $\mathbf{A}$, $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$, and there exist an $m \times m$ orthogonal matrix $\mathbf{U}$ and an $n \times n$ orthogonal matrix $\mathbf{V}$ such that

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T.$$

Note: Any factorization $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$, with $\mathbf{U}$ and $\mathbf{V}$ orthogonal, $\mathbf{\Sigma}$ in the required form, and positive diagonal entries in $\mathbf{D}$, is called a SVD of $\mathbf{A}$. The matrices $\mathbf{U}$ and $\mathbf{V}$ are not uniquely determined by $\mathbf{A}$, but the diagonal entries of $\mathbf{\Sigma}$ are necessarily the singular values of $\mathbf{A}$.

Proof of the Existence

Proof: Let λ_i and $\mathbf{v}_i$ be the eigenvalue and the corresponding eigenvector of $\mathbf{A}^T\mathbf{A}$.
Assume there are r nonzero eigenvalues λ_i, denoted as

$$\lambda_1 \geq \lambda_2 \cdots \geq \lambda_r > 0$$

Define

$$\mathbf{u}_i = \frac{1}{||\mathbf{A}\mathbf{v}_i||}\mathbf{A}\mathbf{v}_i = \frac{1}{\sigma_i}\mathbf{A}\mathbf{v}_i,$$

and

$$\mathbf{A}\mathbf{v}_i = \sigma_i \mathbf{u}_i, \text{ for } 1 \leq i \leq r.$$

Now extend $\{\mathbf{u}_1\ \mathbf{u}_2\ \cdots\ \mathbf{u}_r\}$ to an orthonormal basis $\{\mathbf{u}_1\ \mathbf{u}_2\ \cdots\ \mathbf{u}_m\}$ of $\mathbb{R}^m$ (self-study the Gram-Schmidt process). Let

$$\mathbf{U} = [\mathbf{u}_1\ \mathbf{u}_2\ \cdots\ \mathbf{u}_m] \text{ and } \mathbf{V} = [\mathbf{v}_1\ \mathbf{v}_2\ \cdots\ \mathbf{v}_n]$$

By construction, $\mathbf{U}$ and $\mathbf{V}$ are orthogonal matrices. Also,

$$\mathbf{AV} = [\mathbf{Av}_1 \ \cdots \ \mathbf{Av}_r \ \mathbf{0} \ \cdots \ \mathbf{0}] = [\sigma_1 \mathbf{u}_1 \ \cdots \ \sigma_r \mathbf{u}_r \ \mathbf{0} \ \cdots \ \mathbf{0}]$$

Let $\mathbf{D}$ be the diagonal matrix with diagonal entries $\sigma_1, \cdots, \sigma_r$, and let $\mathbf{\Sigma} = \begin{bmatrix} \mathbf{D} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$.
Then

$$\mathbf{U\Sigma} = [\mathbf{u}_1 \ \mathbf{u}_2 \ \cdots \ \mathbf{u}_m] \begin{bmatrix} \mathbf{D} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} = [\sigma_1 \mathbf{u}_1 \ \cdots \ \sigma_r \mathbf{u}_r \ \mathbf{0} \ \cdots \ \mathbf{0}] = \mathbf{AV}$$

Since $\mathbf{V}$ is an orthogonal matrix, $\mathbf{U\Sigma V}^T = \mathbf{AVV}^T = \mathbf{A}$.

Example 5

Construct a singular value decomposition of $\mathbf{A} = \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix}$.

Solution:

1. **Find an orthogonal diagonalization of $\mathbf{A}^T\mathbf{A}$, i.e., $\mathbf{A}^T\mathbf{A} = \mathbf{PBP}^{-1}$** Clearly,

$$\mathbf{P} = [\mathbf{v}_1\ \mathbf{v}_2\ \mathbf{v}_3] = \begin{bmatrix} 1/3 & -2/3 & 2/3 \\ 2/3 & -1/3 & -2/3 \\ 2/3 & 2/3 & 1/3 \end{bmatrix} \text{ and } \mathbf{B} = \begin{bmatrix} 360 & 0 & 0 \\ 0 & 90 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

2. **Set up V and $\mathbf{\Sigma}$**

$\mathbf{V} = \mathbf{P}$ and $\mathbf{\Sigma} = \begin{bmatrix} \sqrt{360} & 0 \\ 0 & \sqrt{90} \end{bmatrix}$

Hence,

$$\mathbf{\Sigma} = \begin{bmatrix} \sqrt{360} & 0 & 0 \\ 0 & \sqrt{90} & 0 \end{bmatrix}$$

3 **Construct** U

$$\mathbf{u}_1 = \frac{1}{\sigma_1}\mathbf{A}\mathbf{v}_1 = \frac{1}{\sqrt{360}}\begin{bmatrix} 18 \\ 6 \end{bmatrix} = \begin{bmatrix} 3/\sqrt{10} \\ 1/\sqrt{10} \end{bmatrix}$$

$$\mathbf{u}_2 = \frac{1}{\sigma_2}\mathbf{A}\mathbf{v}_2 = \frac{1}{\sqrt{90}}\begin{bmatrix} 3 \\ -9 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{10} \\ -3/\sqrt{10} \end{bmatrix}$$

Since $\{\mathbf{u}_1, \mathbf{u}_2\}$ is already a basis for $\mathbb{R}^2$, no additional vectors are needed for $\mathbf{U}$, and $\mathbf{U} = [\mathbf{u}_1\ \mathbf{u}_2]$.

Verify that

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$$

$$= \begin{bmatrix} 3/\sqrt{10} & 1/\sqrt{10} \\ 1/\sqrt{10} & -3/\sqrt{10} \end{bmatrix} \begin{bmatrix} \sqrt{360} & 0 & 0 \\ 0 & \sqrt{90} & 0 \end{bmatrix} \begin{bmatrix} 1/3 & 2/3 & 2/3 \\ -2/3 & -1/3 & 2/3 \\ 2/3 & -2/3 & 1/3 \end{bmatrix}$$

Example 6

Construct a singular value decomposition of $\mathbf{A} = \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix}$.

Solution:

1. $\mathbf{A}\mathbf{A}^T = \begin{bmatrix} 9 & -9 \\ -9 & 9 \end{bmatrix}$. The eigenvalues 18 and 0, with corresponding unit eigenvectors

$$\mathbf{v}_1 = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

2. Hence, $\mathbf{V} = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$.

   The singular values are $\sigma_1 = \sqrt{18} = 3\sqrt{2}$ and $\sigma_2 = 0$. Since only $\sigma_1 > 0$, $\mathbf{D}$ will be a single number, i.e., $\mathbf{D} = 3\sqrt{2}$.
   Now,

$$\mathbf{\Sigma} = \begin{bmatrix} 3\sqrt{2} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

3 To construct $\mathbf{U}$, we have

$$\mathbf{A}\mathbf{v}_1 = \begin{bmatrix} 2/\sqrt{2} \\ -4/\sqrt{2} \\ 4/\sqrt{2} \end{bmatrix}, \ \mathbf{A}\mathbf{v}_2 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Hence,

$$\mathbf{u}_1 = \frac{1}{3\sqrt{2}}\mathbf{A}\mathbf{v}_1 = \begin{bmatrix} 1/3 \\ -2/3 \\ 2/3 \end{bmatrix}$$

The other columns of $\mathbf{U}$ are found by extending the set $\{\mathbf{u}_1\}$ to an orthonormal basis for $\mathbb{R}^3$. We need two orthogonal unit vectors $\mathbf{u}_2$ and $\mathbf{u}_3$ that are orthogonal to $\mathbf{u}_1$.
Let $\mathbf{x}^T = [x_1 \ x_2 \ x_3]$, if $\mathbf{x}$ and $\mathbf{u}_1$ are orthogonal, then $\mathbf{u_1}^T\mathbf{x} = 0$ yielding $x_1 - 2x - 2 + 2x_3 = 0$. So a basis for the solution set of this equation is

$$\mathbf{w}_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \ \mathbf{w}_2 = \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$$

Apply the Gram-Schmidt process to $\{\mathbf{w}_1, \mathbf{w}_2\}$, and obtain

$$\mathbf{u}_2 = \begin{bmatrix} 2/\sqrt{5} \\ 1/\sqrt{5} \\ 0 \end{bmatrix}, \ \mathbf{u}_3 = \begin{bmatrix} -2/\sqrt{45} \\ 4/\sqrt{45} \\ 5/\sqrt{45} \end{bmatrix}$$

Finally, set $\mathbf{U} = [\mathbf{u}_1 \ \mathbf{u}_2 \ \mathbf{u}_3]$.
Verify that

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$$

Python

- Operations
- Iteration
- Error Handling
- Numpy
- Matrix Operation
- Practices

Operations

Operations

- addition: +
- substraction: -
- multiplication: *
- division: /
- power: **
- modulo: %
- floor division: //

Example

```
(((6+5)-4)*3/2)**2%4
```

2.25

Examples

```
x1 = 1
x2 = 2
x3 = (x1 * x2) - (x1 + x2)
x4 = x3/3
x5 = x3//3

x3
x4
round(x4,2)
x5
```

-1
-0.3333333333333333
-0.33
-1

Iteration

```
for i in range(3):
    print(i)
```

0
1
2

```
print(range(3))
```

range(0, 3)

```
y = [0] * 3
for i in range(3):
    y[i]=i*2
    print(y)
```

$[0, 0, 0]$
$[0, 2, 0]$
$[0, 2, 4]$

```python
for i in range(5, 13, 3):
    print(i)
```

5
8
11

```python
list(range(5))
```

[0, 1, 2, 3, 4]

```python
sum(range(5))
```

10

```python
for c in ['A', 'B', 'C']:
    print(c)
```

A
B
C

Error Handling

```
for d in [2,1,0]:
    print(1/d)
```

0.5
1
ZeroDivisionError

```
for d in [2,1,0]:
    try:
        print(1/d)
    except ZeroDivisionError:
        print("Denominator is 0")
```

0.5
1
Denominator is 0

Numpy

```python
import numpy as np
x = np.array([[1,2,3], [4,5,6], [7,8,9]])

x
x[0] # 0-th row of x
x[0,:] # 0-th row of x
x[1,2] #  (1+1)-th to (2+1)th column of x
```

```
array([[1, 2, 3],
       [4, 5, 6],
       [7, 8, 9]])
array([[1, 2, 3])
array([[1, 2, 3])
6
```

```python
import numpy as np
x = np.array([[1,2,3], [4,5,6], [7,8,9]])

x[:,0] #  0-th column of x
x[:,1:3] #  1-st to (3-1)th column of x
x.sum(axis=0) # axis=0 by column
x.sum(axis=1) # axis=1 by row
```

```
array([[1, 4, 7])
array([[2, 3],
       [5, 6],
       [8, 9]])
array([12, 15, 18])
array([6, 15, 24])
```

Matrix

```python
import numpy as np
x1 = np.array([[1,2], [3,5]])
x2 = np.array([[2],[3]])

np.dot(x1,x2) # dot product
x1.T # transpose
np.linalg.inv(x1) # inverse
np.linalg.det(x1) # determinant
```

```
array([[ 8],
  [21]])
array([[1,3],
       [2,5]])
array([[-5,2],
       [3,-1]])
-1
```

```python
import numpy as np
x1 = np.array([[1,2], [3,5]])

np.linalg.eig(x1) # eigenvalue and eigenvectors
```

(array([-0.16227766, 6.16227766]),# eigenvalues
array([[-0.86460354, -0.36126098],
[0.50245469, -0.93246475]])) # eigenvectors

Practices

- Verify Examples 1-6 by Python!
- Verify results in page 45 and use these results to diagonalize the matrix $\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$
- Construct a SVD of $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 5 \\ 4 & 6 \end{bmatrix}$